

Cardiovascular Outcomes among New Users of GLP-1 Receptor Agonists Compared with DPP-4 Inhibitors and Sulfonylureas in Kidney Failure

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Key Points

- Glucagon-like peptide-1 receptor agonists are increasingly used in patients with kidney failure despite trial exclusion.
- Initiation was associated with a 13% lower risk of cardiovascular events and 17% lower mortality versus dipeptidyl peptidase-4 inhibitors.
- Initiation was associated with a 10% lower risk of heart failure hospitalization versus dipeptidyl peptidase-4 inhibitors.

Abstract

Background Few therapies improve cardiovascular outcomes for people with kidney failure. Glucagon-like peptide-1 receptor agonists (GLP-1 RA) reduce cardiovascular risk in patients with non-dialysis-dependent CKD, but the cardiovascular benefits in patients with kidney failure remain uncertain. The objective of this study was to compare cardiovascular outcomes among patients with kidney failure and type 2 diabetes newly initiated on GLP-1 RA versus other antidiabetic agents.

Methods We analyzed electronic health records, Medicare claims, and Part D data from the United States Renal Data System (2011–2021) to identify new users of GLP-1 RA ($n=3629$), dipeptidyl peptidase-4 inhibitors (DPP4i; $n=21,369$), and sulfonylureas ($n=32,296$) among patients with type 2 diabetes receiving maintenance dialysis. For the primary analysis, we performed 1:1 propensity score matching of GLP-1 RA to DPP4i initiators using 61 covariates. A prespecified secondary analysis compared propensity score-matched initiators of GLP-1 RA and sulfonylureas. The primary outcome was a modified major adverse cardiovascular event (MACE) composite of myocardial infarction, stroke, or all-cause mortality. Secondary outcomes included the individual components of the primary outcome and hospitalizations for heart failure. Cause-specific Cox models were used to estimate hazard ratios (HRs).

Results Among 3284 matched pairs of GLP-1 RA and DPP4i initiators, GLP-1 RA use was associated with lower risks of MACE (HR, 0.87; 95% confidence interval [CI], 0.78 to 0.97), all-cause mortality (HR, 0.83; 95% CI, 0.74 to 0.94), and heart failure hospitalization (HR, 0.90; 95% CI, 0.83 to 0.99) over up to 2 years of follow-up. Among 2792 matched pairs, GLP-1 RA and sulfonylurea initiators, GLP-1 RA was associated with lower risks of MACE (HR, 0.83; 95% CI, 0.74 to 0.93) and all-cause mortality (HR, 0.80; 95% CI, 0.69 to 0.91).

Conclusions Among patients with type 2 diabetes receiving maintenance dialysis, GLP-1 RA initiation was associated with lower risk of cardiovascular events and all-cause mortality compared with other commonly prescribed antidiabetic agents.

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Introduction

Cardiovascular disease is the leading cause of death among people with kidney failure.¹ While select therapies reduce cardiovascular events in patients with non-dialysis-dependent CKD, randomized clinical trials have not consistently demonstrated similar benefits in the

kidney failure population.^{2–4} This disparity likely reflects the unique pathophysiology of cardiovascular disease in kidney failure that is characterized by severe vascular calcifications, disordered mineral metabolism, uremic toxins, chronic inflammation, volume overload, and hemodynamic instability.^{5–7} Furthermore, several emerging

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therapies with proven cardiovascular benefit in earlier stages of CKD remain largely untested in people receiving maintenance dialysis.^{8–12}

Glucagon-like peptide-1 receptor agonists (GLP-1 RA) are a particularly promising candidate to improve cardiovascular outcomes in kidney failure. In patients with type 2 diabetes and CKD, GLP-1 RA slow CKD progression, reduce cardiovascular risk, and improve survival.^{8,13} Although landmark trials excluded patients with kidney failure, several GLP-1 RA formulations have favorable pharmacokinetic profiles and do not require dose adjustment in this population.^{14,15} Beyond glycemic control and weight reduction, GLP-1 RA reduce inflammation,¹⁶ improve endothelial function,¹⁷ and modulate autonomic pathways.¹⁸ These mechanisms may be particularly relevant in kidney failure.

Furthermore, a recent large cohort study found that GLP-1 RA use in kidney failure was associated with a 23% lower risk of mortality, modest weight loss, and a higher likelihood of being listed for transplant compared with nonuse.¹⁹ However, this study had notable limitations, including lack of an active comparator group, potential confounding by indication, and absence of cardiovascular outcome data. Thus, whether GLP-1 RA improve cardiovascular outcomes in kidney failure remains unknown.

To address this critical gap, we analyzed data from the United States Renal Data System (USRDS) to compare cardiovascular outcomes and survival among patients with type 2 diabetes and kidney failure who initiated GLP-1 RA versus either dipeptidyl peptidase-4 inhibitors (DPP4i) or sulfonylureas.

Methods

Study Population

We collected individual patient data from the USRDS, a national registry that captures demographic, clinical, and Medicare Parts A, B, and D claims data for most patients who initiate dialysis in the United States. The study population consisted of adult patients (aged 18–100 years) with kidney failure receiving maintenance hemodialysis or peritoneal dialysis and a diagnosis of type 2 diabetes, identified using International Classification of Diseases, Ninth and Tenth Revisions (ICD-9 and ICD-10) codes, at any time during the study period before the index date ([Supplemental Table 1](#)). We excluded patients with type 1 diabetes and those who died in 2011 (the first year of data collection).

Exposure and Censoring

The primary exposure was the initial prescription of GLP-1 RA, DPP4i, or sulfonylurea between January 1, 2011, and December 31, 2021. We started the exposure window in 2011 to ensure contemporary and overlapping use of these medications. The index date was defined as the date of the first prescription for the medication of interest that occurred after the onset of dialysis. Follow-up data were available through December 31, 2021. Participants were excluded if they were prescribed one of these medications 12 months before the index date to limit the risk of carryover or if they did not receive a repeat prescription at

least 90 days after the index date. The primary analysis included a propensity score–matched cohort of patients who initiated either a GLP-1 RA or DPP4i after starting dialysis. In a prespecified secondary analysis, we separately evaluated a propensity score–matched cohort of new initiators of GLP-1 RA versus sulfonylurea. We censored for follow-up at 2 years, kidney transplantation, or initiation of a competing therapy (e.g., GLP-1 RA initiator subsequently prescribed DPP4i and *vice versa*).

Outcomes

The primary outcome was a modified major adverse cardiovascular event (MACE) composite of nonfatal myocardial infarction, nonfatal stroke, and all-cause mortality. Secondary outcomes included nonfatal cardiovascular events (myocardial infarction and stroke), heart failure hospitalization, and all-cause mortality. Outcomes were assessed as time-to-first event. Cardiovascular events were based on ICD-9 and ICD-10 codes available in the USRDS institutional claims file ([Supplemental Table 1](#)). Death events were captured from the death file or the patient profile file within USRDS ([Supplemental Table 2](#)). Because hospitalization events for noncardiac events in the kidney failure population are common, we focused on cardiovascular events rather than all-cause hospitalizations. Finally, because cause-specific mortality is subject to misclassification in USRDS,²⁰ we analyzed all-cause mortality.

Covariates

Covariates used to estimate the propensity scores were selected on the basis of their potential association with either the exposure (*i.e.*, choice of antidiabetic medication) or outcomes. Covariate data were assessed on the index date or on the closest available time point before it. [Supplemental Table 2](#) lists the source of each covariate. Key covariates included demographic characteristics, comorbidities, year of prescription, concomitant medications, body mass index, select dialysis laboratory values (albumin and hemoglobin), primary cause of kidney failure, dialysis modality, dialysis vintage, kidney transplant waitlist status, tobacco use, and the number of hospitalizations in the preceding 12 months.

Statistical Analysis

Baseline characteristics were reported for the overall cohort and the matched cohorts comparing GLP-1 RA initiators with DPP4i initiators and sulfonylurea initiators. The primary analysis compared patients initiating GLP-1 RA versus DPP4i, while a prespecified secondary analysis compared GLP-1 RA initiators with sulfonylurea initiators.

Propensity scores were estimated using a multistep process. First, missing data were addressed using multiple imputation with classification and regression trees, generating 20 complete datasets.²¹ For each dataset, least absolute shrinkage and selection operator logistic regression was applied to model the logit-transformed probability of a patient initiating on a GLP-1 RA using 61 covariates ([Supplemental Table 2](#)).²² Main effects were forced into the model, while interaction terms between sex, race, and

age categories with other covariates were included for variable selection. Three-fold cross-validation was used to select the optimal regularization parameter. Propensity scores for each participant were then averaged across the 20 imputed datasets. We constructed matched cohorts using 1:1 nearest neighbor matching with a caliper of 0.2 and exact matching on the year of drug initiation. Covariate balance was assessed using the standardized mean differences (SMD) for continuous characteristics and absolute differences in proportion for categorical variables.

Imputation, propensity score estimation, and matching were performed separately for each comparison (*i.e.*, GLP-1 RA versus DPP4i and GLP-1 RA versus sulfonylureas). For myocardial infarctions, strokes, and heart failure hospitalizations, we considered death as a competing risk. Cause-specific hazards were estimated using the Cox proportional hazards models, and cumulative incidence functions were modeled using the Fine–Gray subdistribution hazard model.^{23,24} The Fine–Gray model was used as a sensitivity analysis to the cause-specific hazards modeling. Outcomes without a competing risk (*i.e.*, MACE and all-cause mortality) were assessed using standard Cox models. Factors that remained imbalanced after propensity score matching (*i.e.*, SMD >0.1) were included as covariates in the Cox models to adjust for residual differences between groups.

The proportional hazards assumption was assessed in Cox models using the test developed by Grambsch and Therneau, with a *P* value of 0.05 serving as the cutoff for significance.²⁵ Time-varying models were used when violations were detected. In Fine–Gray models, visual inspection of Schoenfeld residuals was used to identify covariates that potentially violated the proportional hazards assumption. For these covariates, transformations of time (*i.e.*, linear, quadratic, and log) were applied and tested for significance. A *P* value cutoff of 0.05 was used to determine significance for all covariates. *R* version 4.4.2 was used for all analyses.

By conducting separate propensity score–matched comparisons restricted to new users of active comparators, we minimize confounding by indication and reduce channeling bias. We evaluated residual confounding using time to first vascular access intervention as a falsification end point because it is unlikely to be influenced by the studied treatments. We defined the index date as the first qualifying prescription for the target antidiabetic agent to mitigate immortal time bias, and we censored follow-up at 2 years to limit the influence of differential follow-up duration. This design also improves covariate balance by reducing within-comparator heterogeneity and supports more valid outcome ascertainment than landmark-based approaches by naturally aligning index dates for GLP-1 RA and active comparators.

Sample Size

We estimated that a minimum of 1600 patients (800 propensity score–matched pairs) would be required to detect a 20% relative risk reduction in the primary outcome, assuming an incidence rate of 194 MACE events per 1000 patient-years in the referent group over an average of 2 years of follow-up, 80% power, and an α of 0.05.²⁶

Results

Study Population and Baseline Characteristics

A total of 108,049 patients within USRDS received maintenance dialysis with at least two fills of a GLP-1 RA or a DPP4i between January 1, 2011, and December 31, 2021. After applying the exclusion criteria, 3629 patients initiated GLP-1 RA, and 21,369 initiated DPP4i. Ultimately, the propensity score–matched cohort for the primary analysis included 6568 patients, with 3284 patients initiating a GLP-1 RA and 3284 initiating a DPP4i (Figure 1). The most prescribed GLP-1 RA was liraglutide (Supplemental Table 6). Baseline characteristics before and after propensity score matching are presented in Table 1. The median age of the matched cohort was 62 (interquartile range, 53–69) years; 49% of patients were women, and 27% identified as Black. Overall, 85% of the matched cohort were classified as overweight or obese. Most patients (89%) were receiving hemodialysis. Covariate balance was achieved across all variables (SMD <0.1), except for dialysis vintage, which was slightly longer among DPP4i initiations (3.5 years versus 3.1 years; SMD=0.12). Thus, all analyses were adjusted for dialysis vintage.

Clinical Outcomes

The median time from index date to the primary outcome of MACE or end of follow-up was 222 (interquartile range, 148–300) days among GLP-1 RA initiators and 245 (interquartile range, 151–435) days among DPP4i initiators. Medication discontinuation was the most common reason for censoring in both groups. The primary outcome of MACE occurred in 19% of GLP-1 RA initiators (243.9 events per 1000 patient-years) and 24% of DPP4i initiators (276.9 events per 1000 patient-years). GLP-1 RA initiation was associated with a significantly lower risk of MACE compared with DPP4i initiation (hazard ratio [HR], 0.87; 95% confidence interval [CI], 0.78 to 0.97; Figure 2 and Table 2). Risks for the individual components of MACE followed a similar pattern: The HR for nonfatal myocardial infarction or stroke was 0.88 (95% CI, 0.75 to 1.04), and the HR for all-cause mortality was 0.83 (95% CI, 0.74 to 0.94), with only mortality reaching statistical significance. Among the components of the primary outcome, the incidence of mortality among GLP-1 RA initiators was 171.3 per 1000 patient-years, compared with 203.0 among DPP4i initiators. Nonfatal myocardial infarctions occurred at 78.2 events per 1000 patient-years in the GLP-1 RA group as compared with 84.2 in the DPP4i group. Nonfatal strokes occurred at a rate of 25.7 events per 1000 patient-years in the GLP-1 RA group as compared with 29.7 events in the DPP4i group. Finally, GLP-1 RA initiation was associated with a significantly lower risk of heart failure hospitalization (HR, 0.90; 95% CI, 0.83 to 0.99).

Subgroup Analysis

Subgroup analysis for the primary outcome revealed significant effect modification by sex (*P* for interaction=0.04) and history of cardiovascular disease (*P* for interaction=0.05; Figure 3). Among women, GLP-1 RA initiation was associated with a more pronounced MACE reduction compared with men, a pattern observed for both mortality and cardiovascular events (Supplemental Figures 1 and 2).

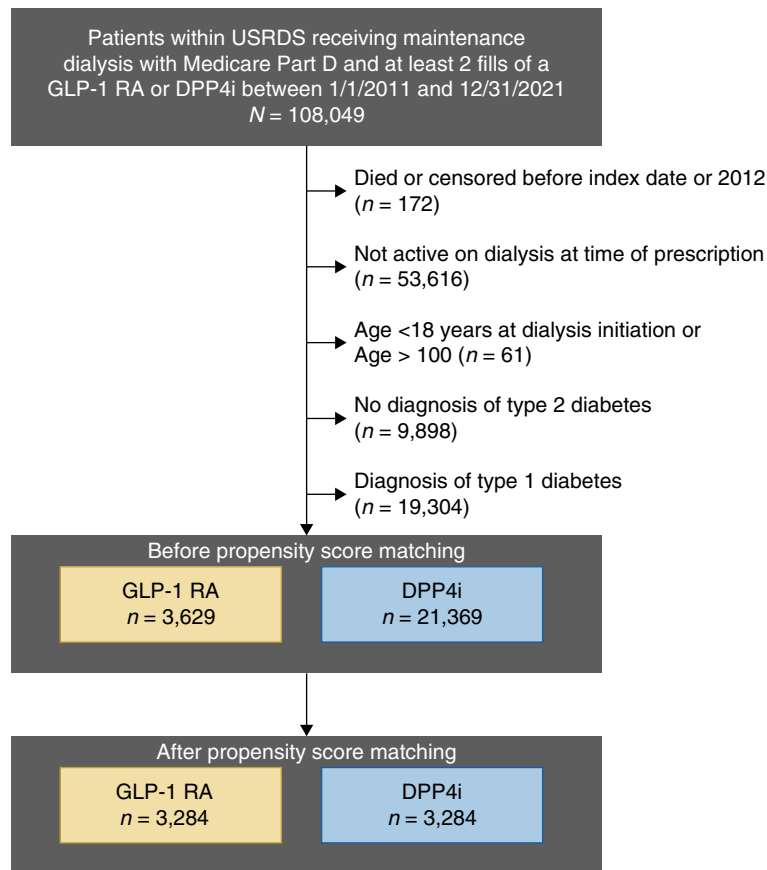


Figure 1. Consort diagram: GLP-1 RA initiators as compared with the DPP4i initiators. DPP4i, dipeptidyl peptidase-4 inhibitor; GLP-1 RA, glucagon-like peptide 1 receptor agonist; USRDS, United States Renal Data System.

In the subgroup analysis of patients with a history of cardiovascular disease, stratified analyses were conducted to account for nonproportional hazards. During the early follow-up period (0–8 months), GLP-1 RA initiation was associated with a significantly lower risk of MACE in patients with a history of cardiovascular disease, although this association attenuated later in follow-up (Figure 3). No effect modification was observed for age, body mass index, history of heart failure, or dialysis vintage. Subgroup analysis for the heart failure hospitalization is reported in Supplemental Figure 3.

Secondary Analysis

The propensity score-matched cohort for the secondary analysis of GLP-1 RA versus sulfonylurea users comprised 5584 patients (2792 in each group; Supplemental Figure 4). Baseline characteristics were similar to those in the primary cohort, with good balance achieved (Supplemental Table 3). GLP-1 RA initiation was associated with significantly lower risk of MACE compared with sulfonylurea initiation (HR, 0.83; 95% CI, 0.74 to 0.93). Risks for the individual components of MACE followed a similar pattern: The HR for nonfatal myocardial infarction or stroke was 0.83 (95% CI, 0.68 to 1.00), and the HR for all-cause mortality was 0.80 (95% CI, 0.69 to 0.91; Table 2 and Supplemental Figure 5). The risk of heart failure hospitalization did not significantly differ between

groups (HR, 0.93; 95% CI, 0.85 to 1.03). Subgroup analyses are reported in Supplemental Figure 6.

Sensitivity Analyses

Competing risk analyses of individual cardiovascular outcomes using Fine-Gray subdistribution hazard models yielded results consistent with the primary analysis. For the primary cohort, the association between GLP-1 RA initiation and nonfatal myocardial infarction and stroke was similar to cause-specific hazards modeling (HR, 0.89; 95% CI, 0.75 to 1.03 compared with DPP4i initiation; Supplemental Table 4). The association between GLP-1 RA initiation and heart failure hospitalization was consistent with the cause-specific hazards model (HR, 0.91; 95% CI, 0.82 to 0.98). Findings from the secondary cohort were similar between the Fine-Gray and cause-specific hazard models (Supplemental Table 4). No association was observed between GLP-1 RA initiation and the falsification end point of access intervention in either cohort (Supplemental Table 5), suggesting that unmeasured confounding was unlikely to fully account for the observed associations with survival and cardiovascular outcomes.

Discussion

In this large, national study of patients with kidney failure, GLP-1 RA initiation was associated with a lower risk of

Table 1. Baseline characteristics of the glucagon-like peptide-1 receptor agonist initiators as compared with the dipeptidyl peptidase-4 inhibitor initiators before and after propensity score matching

Characteristics	Overall (N=24,998)	Unmatched			Matched		
		GLP-1 RA (n=3629)	DPP4i (n=21,369)	SMD	GLP-1 RA (n=3284)	DPP4i (n=3284)	SMD
Age, yr, n (%)							
Younger than 50	2647 (11)	725 (20)	1922 (9)	0.11	589 (18)	562 (17)	0.008
50–59	4940 (20)	980 (27)	3960 (19)	0.085	869 (26)	828 (25)	0.012
60–69	7973 (32)	1175 (32)	6798 (32)	0.006	1090 (33)	1106 (34)	0.005
70 or older	9438 (38)	749 (21)	8689 (41)	0.2	726 (22)	788 (24)	0.016
Sex, n (%)							
Male	13,243 (53)	1833 (51)	11,410 (53)		1667 (51)	1712 (52)	
Female	11,755 (47)	1796 (49)	9959 (47)	0.029	1617 (49)	1572 (48)	0.027
Race, n (%)							
Asian	1694 (7)	173 (5)	1521 (7)	0.024	161 (4)	157 (5)	0.001
Black	6950 (28)	991 (27)	5959 (28)	0.006	883 (27)	889 (27)	0.002
Other	739 (3)	123 (3)	616 (3)	0.005	112 (3)	107 (3)	0.002
White	15,615 (62)	2342 (65)	13,273 (62)	0.024	2128 (65)	2131 (65)	0.001
Ethnicity, n (%)							
Hispanic	5815 (23)	777 (21)	5038 (24)	0.021	722 (22)	764 (23)	0.013
BMI, kg/m², n (%)							
Obesity class 3 (≥40 kg/m ²)	2923 (12)	1005 (28)	1918 (9)	0.187	799 (24)	660 (20)	0.043
Obesity class 2 (35–39.9 kg/m ²)	2981 (12)	705 (19)	2276 (11)	0.085	625 (19)	606 (18)	0.006
Obesity class 1 (30–34.9 kg/m ²)	5026 (20)	858 (24)	4168 (20)	0.033	805 (25)	815 (25)	0.003
Overweight (25–29.9 kg/m ²)	6787 (27)	619 (17)	6168 (29)	0.134	615 (19)	677 (21)	0.02
Normal and underweight (≤24.9 kg/m ²)	5836 (23)	365 (10)	5471 (26)	0.171	365 (11)	447 (14)	0.026
Region^a, n (%)							
Midwest	4238 (17)	680 (19)	3558 (17)	0.021	612 (19)	588 (18)	0.007
Northeast	4837 (19)	522 (14)	4315 (20)	0.058	488 (15)	518 (16)	0.009
South	10,250 (41)	1613 (44)	8637 (40)	0.04	1438 (44)	1433 (44)	0.002
West	5673 (23)	814 (22)	4859 (23)	0.003	746 (23)	745 (23)	0
Primary cause of kidney failure, n (%)							
Diabetes	18,639 (75)	2836 (78)	15,803 (74)	0.044	2556 (78)	2524 (77)	0.012
Hypertension	4088 (16)	484 (13)	3604 (17)	0.035	446 (14)	469 (14)	0.007
Other	1072 (4)	140 (4)	932 (4)	0.005	127 (4)	146 (5)	0.006
GN	539 (2)	83 (2)	456 (2)	0.002	76 (2)	67 (2)	0.003
Unknown, other urologic, cystic ^b	561 (2)	64 (2)	497 (2)		57 (2)	64 (2)	
Predominant dialysis modality, n (%)							
PD	1947 (8)	366 (10)	1581 (7)		328 (10)	314 (10)	
Hemodialysis	22,569 (90)	3195 (88)	19,374 (91)	0.027	2894 (88)	2921 (89)	0.005
Kt/V (mean)							
Hemodialysis	1.6 (0.3)	1.6 (0.3)	1.6 (0.3)	0.113	1.6 (0.3)	1.6 (0.3)	0.029
PD	2.2 (0.6)	2.3 (0.5)	2.2 (0.6)	0.084	2.3 (0.5)	2.2 (0.6)	0.085
Waitlist, kidney transplant, n (%)							
Yes	2697 (11)	476 (13)	2221 (10.39)	0.027	428 (13)	404 (12)	0.007
No	22,301 (89)	3153 (87)	19,148 (90)		2856 (87)	2880 (88)	
Dialysis vintage, yr							
Mean (SD)	2.73 (3.43)	3.22 (3.51)	2.64 (3.41)	0.164	3.11 (3.44)	3.53 (4.02)	0.121
Median (IQR)	1.52 (0.33–3.84)	2.20 (0.81–4.47)	1.38 (0.29–3.71)		2.08 (0.74–4.34)	2.23 (0.77–4.88)	
Type of access, n (%)							
Arteriovenous fistula	11,774 (47)	1938 (53)	9836 (46)	0.085	1736 (53)	1772 (54)	0.003
Arteriovenous graft	3311 (13)	535 (15)	2776 (13)	0.02	472 (14)	475 (14)	0.002
Central venous catheter	6734 (27)	722 (20)	6012 (28)	0.09	675 (21)	689 (21)	0.001
Multiple/other/ unknown ^b	667 (3)	54 (1)	613 (3)		51 (2)	45 (1)	
Prior transplant (yes), n (%)	465 (2)	77 (2)	388 (2)	0.003	71 (2)	87 (3)	0.005
Comorbidities, n (%)							
Atrial fibrillation	4624 (19)	600 (17)	4024 (19)	0.023	540 (16)	621 (19)	0.025
Other cardiac dysrhythmias	5824 (23)	822 (23)	5002 (23)	0.008	740 (23)	855 (26)	0.035

Table 1. Continued

Characteristics	Overall (N=24,998)	Unmatched			Matched		
		GLP-1 RA (n=3629)	DPP4i (n=21,369)	SMD	GLP-1 RA (n=3284)	DPP4i (n=3284)	SMD
Hypertension	23,643 (95)	3500 (96)	20,143 (94)	0.022	3161 (96)	3194 (97)	0.01
Atherosclerosis	13,916 (56)	1956 (54)	11,960 (56)	0.021	1789 (54)	1883 (57)	0.029
Pericarditis	3690 (15)	564 (16)	3126 (15)	0.009	506 (15)	537 (16)	0.009
COPD	3955 (16)	680 (19)	3275 (15)	0.034	594 (18)	665 (20)	0.022
Dementia	1650 (7)	132 (4)	1518 (7)	0.035	126 (4)	130 (4)	0.001
Congestive heart failure	13,859 (55)	2045 (56)	11,814 (55)	0.011	1835 (56)	1935 (59)	0.03
Peripheral vascular disease	6838 (27)	899 (25)	5939 (28)	0.03	828 (25)	848 (26)	0.006
HIV/AIDS	218 (0.87)	34 (1)	184 (0.86)	0.001	31 (1)	35 (1)	0.001
Amputations	841 (3)	175 (5)	666 (3)	0.017	151 (5)	155 (5)	0.001
Solid tumors	829 (3)	94 (3)	735 (3)	0.008	85 (3)	121 (4)	0.011
Lymphoma and leukemia ^b	248 (0.99)	22 (0.61)	226 (1)		20 (0.6)	17 (0.5)	
Liver disease	2399 (10)	305 (8)	2094 (10)	0.014	276 (8)	301 (9)	0.008
Hyperlipidemia	14,177 (57)	2273 (63)	11,904 (56)	0.069	2007 (61)	2161 (66)	0.047
Connective tissue disease	403 (2)	73 (2)	330 (2)	0.005	64 (2)	62 (2)	0.001
Thyroid gland disorders	4174 (17)	647 (18)	3527 (17)	0.013	579 (18)	628 (19)	0.015
Diabetes complications: neuropathy	8209 (33)	1538 (42)	6671 (31)	0.112	1330 (41)	1422 (43)	0.028
Diabetic retinopathy	6395 (26)	1101 (30)	5294 (25)	0.056	972 (30)	1041 (32)	0.021
Diabetes complications: peripheral vascular disease	6238 (25)	1001 (28)	5237 (25)	0.031	869 (26)	980 (30)	0.034
Diabetic foot ulcer	2958 (12)	564 (16)	2394 (11)	0.043	476 (14)	558 (17)	0.025
Current smoker	953 (4)	155 (4)	798 (4)	0.003	135 (4)	142 (4.32)	0.002
History of alcohol dependence	623 (2)	79 (2)	544 (3)	0.004	72 (2)	88 (3)	0.005
History of drug dependence	932 (3)	180 (5)	752 (4)	0.014	149 (5)	189 (5.76)	0.012
Medication use, n (%)							
ESA	23,292 (93)	3475 (96)	19,817 (93)	0.03	3142 (96)	3178 (97)	0.011
Insulin	9536 (38)	2487 (69)	7049 (33)	0.355	2153 (66)	2121 (65)	0.01
ACEi or ARB	9884 (40)	1388 (38)	8496 (40)	0.015	1276 (39)	1297 (39)	0.006
β blocker	16,227 (65)	2511 (69)	13,716 (64)	0.05	2263 (69)	2292 (70)	0.009
Thiazide diuretic	1627 (7)	300 (8)	1327 (6)	0.021	262 (8)	255 (8)	0.002
Loop diuretic	8424 (34)	1541 (42)	6883 (32)	0.103	1365 (42)	1376 (42)	0.003
CCB	12,815 (51)	1870 (52)	10,945 (51)	0.003	1704 (52)	1826 (56)	0.037
Statin	15,460 (62)	2562 (71)	12,898 (60)	0.102	2290 (70)	2385 (73)	0.029
Other lipid-lowering medication	2123 (8)	418 (12)	1705 (8)	0.035	367 (11)	294 (9)	0.022
Corticosteroid	3400 (14)	629 (17)	2771 (13)	0.044	538 (16)	619 (19)	0.025
Antiplatelet	5416 (21)	852 (23)	4564 (21)	0.021	773 (24)	805 (25)	0.01
DOAC	1366 (5)	342 (9)	1024 (5)	0.046	267 (8)	397 (12)	0.04
Meglitinide	811 (3)	70 (2)	741 (3)	0.015	69 (2)	67 (2)	0.001
Sulfonylurea	5225 (21)	481 (13)	4744 (22)	0.089	462 (14)	466 (14)	0.001
Phosphate binder	12,088 (48)	1917 (53)	10,171 (48)	0.052	1742 (53)	1644 (50)	0.03
Cinacalcet	2501 (10)	2173 (10)	328 (9)	0.011	326 (10)	228 (7)	0.03
Institutionalized	1522 (6)	216 (6)	1306 (6)	0.002	201 (6)	208 (6)	0.002
Number of hospitalizations in last yr	1.08 (1.65)	0.87 (1.40)	1.11 (1.69)	0.176	0.87 (1.43)	0.92 (1.49)	0.029
Number of days spent in ICU in last yr	2.54 (6.77)	1.86 (5.41)	2.66 (6.97)	0.149	1.90 (5.47)	2.28 (6.30)	0.069
Number of inpatient charges in last yr	1.17 (1.81)	0.93 (1.53)	1.21 (1.86)	0.183	0.94 (1.56)	1.00 (1.64)	0.037
Time spent in skilled nursing in last yr	11.34 (32.97)	8.63 (31.54)	11.80 (33.18)	0.101	8.84 (31.64)	10.08 (31.35)	0.039

Table 1. Continued

Characteristics	Overall (N=24,998)	Unmatched			Matched		
		GLP-1 RA (n=3629)	DPP4i (n=21,369)	SMD	GLP-1 RA (n=3284)	DPP4i (n=3284)	SMD
Laboratory tests							
Albumin, g/dl	3.6 (0.5)	3.7 (0.4)	3.6 (0.5)	0.411	3.7 (0.4)	3.8 (0.4)	0.064
Hemoglobin, g/dl	10.4 (1.4)	10.8 (1.3)	10.4 (1.4)	0.32	10.7 (1.3)	10.7 (1.4)	0.014

Values are *n* or mean. Continuous data represent mean with SD unless otherwise noted. ACEI, angiotensin-converting enzyme inhibitor; AIDS, acquired immunodeficiency syndrome; ARB, angiotensin II receptor blocker; BMI, body mass index; CCB, calcium channel blocker; COPD, chronic obstructive pulmonary disease; DOAC, direct oral anticoagulant; DPP4i, dipeptidyl peptidase-4 inhibitor; ESA, erythropoiesis-stimulating agent; GLP-1 RA, glucagon-like peptide-1 receptor agonist; HIV, human immunodeficiency virus; ICU, intensive care unit; IQR, interquartile range; KT/V, dialysis adequacy measure (K is dialyzer clear of urea, T is time, V is urea distribution volume); PD, peritoneal dialysis; SMD, standard mean difference.

^aThe geographic regions were created using the Census Region and Divisions of the United States per the US government.

^bEach of the categories included within this composite group were well matched in the final cohort with absolute standard mean differences values less than the cutoff of 0.1. The exact numbers could not be included because of the United States Renal Data System requirement that all numbers with less than 11 participants be suppressed.

modified MACE, hospitalization for heart failure, and all-cause mortality compared with DPP4i initiation over a 2-year follow-up period. These associations were consistent across comparisons with sulfonylureas, except for heart failure hospitalization. Our findings suggest that the cardiovascular and survival benefits observed with GLP-1 RA in patients with CKD not requiring dialysis may also extend to those with kidney failure receiving dialysis.

The 13%–17% reduction in the risk of modified MACE observed in our study aligns with findings from GLP-1 RA cardiovascular outcomes trials. In a recent meta-analysis of 13 randomized, placebo-controlled clinical trials, including populations with obesity, diabetes, heart failure, and other cardiovascular risk factors, GLP-1 RA reduced MACE by 14% compared with placebo.²⁷ Notably, the magnitude of MACE reduction in our study is consistent with the 18%

reduction demonstrated in the Evaluate Renal Function with Semaglutide Once Weekly (FLOW) trial (Effects of Semaglutide on CKD in Patients with Type 2 Diabetes).⁸ Unlike FLOW, which observed a relatively higher risk of nonfatal stroke with advancing kidney disease severity,⁸ our study showed relatively equal reductions in the risks of nonfatal myocardial infarction and stroke, which may reflect the unique cardiovascular burden of kidney failure. Our reliance on all-cause mortality rather than cardiovascular mortality may have biased our results toward the null because the FLOW trial found a larger relative reduction in cardiovascular death (29%) compared with all-cause mortality (20%) with GLP-1 RA therapy.⁸ Consistent with our findings, a recent national cohort study reported a 23% lower mortality risk among patients with kidney failure who initiated GLP-1 RA compared with other

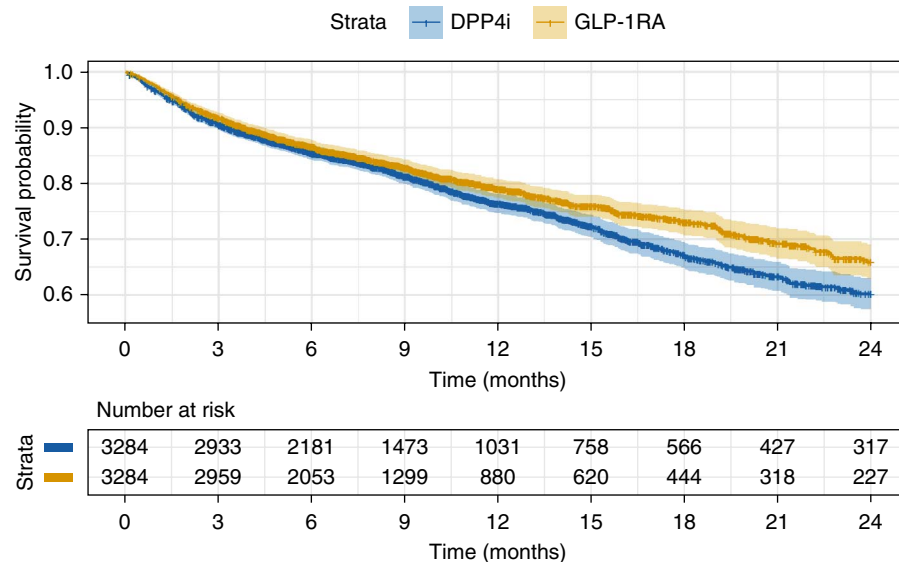


Figure 2. Kaplan–Meier curve for modified MACE in GLP-1 RA initiators versus DPP4i initiators. MACE, major adverse cardiovascular events.

Table 2. Cardiovascular outcomes and survival for glucagon-like peptide-1 receptor agonist initiators as compared with the dipeptidyl peptidase-4 inhibitor initiators and sulfonylurea initiators

Outcome	GLP-1 RA (<i>n</i> =3284) Incidence Rate per 1000 Pt-yrs	DPP4i (<i>n</i> =3284) Incidence Rate per 1000 Pt-yrs	HR (95% CI)	GLP-1 RA (<i>n</i> =2792) Incidence Rate per 1000 Pt-yrs	Sulfonylurea (<i>n</i> =2792) Incidence Rate per 1000 Pt-yrs	HR (95% CI)
Primary outcome (modified MACE)	243.9	276.9	0.87 (0.78 to 0.97)	222.4	267.0	0.83 (0.74 to 0.93)
Components of the primary outcome						
All-cause mortality	171.3	203.0	0.83 (0.74 to 0.94)	156.4	194.5	0.80 (0.69 to 0.91)
Nonfatal MI or stroke	97.1	108.8	0.88 (0.75 to 1.04)	85.9	103.5	0.83 (0.68 to 1.00)
Nonfatal MI	78.2	84.2		64.7	89.9	
Nonfatal stroke	25.7	29.7		20.0	23.5	
Secondary outcome						
Heart failure hospitalization	391.1	420.5	0.90 (0.83 to 0.99)	374.2	396.2	0.93 (0.85 to 1.03)

After propensity score matching, all outcomes models were adjusted for dialysis vintage because of a residual standard mean difference >0.1. CI, confidence interval; DPP4i, dipeptidyl peptidase-4 inhibitor; GLP-1 RA, glucagon-like peptide 1 receptor agonist; HR, hazard ratio; MACE, major adverse cardiovascular events; MI, myocardial infarction.

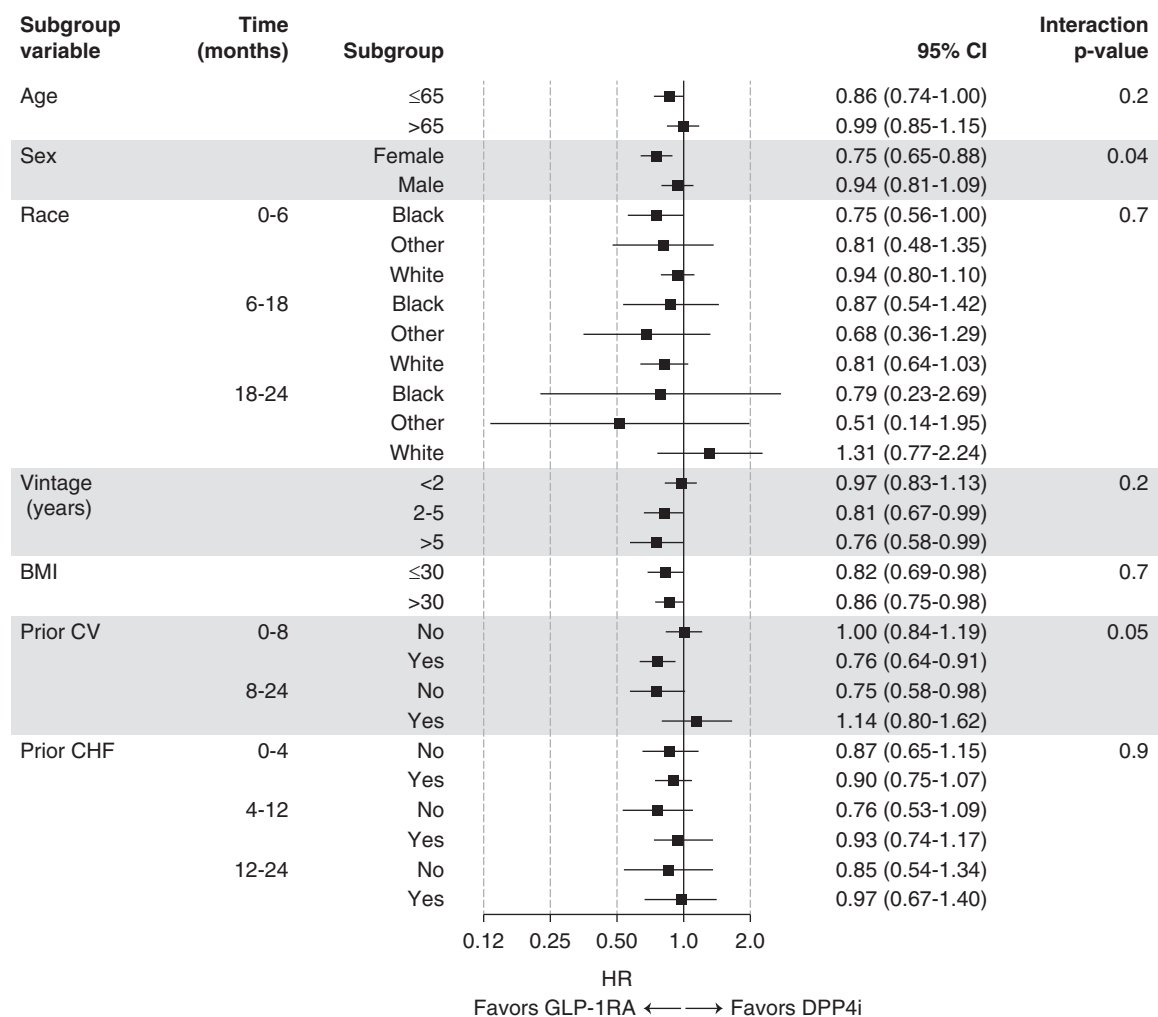


Figure 3. Forest plot comparing subgroups within the GLP-1 RA initiators as compared with the DPP4i initiators analysis. For subgroup analyses where the proportional hazards assumption is violated, the results are further subdivided by follow-up windows presented in months. BMI, body mass index; CHF, congestive heart failure; CI, confidence interval; CV, cardiovascular disease; HR, hazard ratio.

dialysis patients within the USRDS data.¹⁹ Our results also align with the 20% reduction in all-cause mortality observed for semaglutide in the FLOW trial.⁸ Similarly, a comparative effectiveness study of mostly predialysis patients with CKD stage 5 in Taiwan found GLP-1 RA initiation associated with a 21% reduction in all-cause mortality,²⁸ although the small number of GLP-1 RA initiators on hemodialysis (136) limited subgroup analysis. Taken together with the observed cardiovascular benefits, these findings suggest that a substantial component of the mortality reduction associated with GLP-1 RA use in kidney failure is likely driven by reductions in cardiovascular risk.

The risk of heart failure hospitalization was 10% lower for GLP-1 RA initiators compared with DPP4i with a similar magnitude, but nonsignificant association in the comparison with sulfonylurea initiators. Despite the proportionally higher risk of heart failure in kidney failure,²⁹ these data differ from the 27% reduction in heart failure events in the FLOW trial.⁸ This difference may reflect a unique cardiovascular burden of kidney failure, limitations

in event adjudication between study designs, or inherent challenges to heart failure diagnosis in kidney failure. Furthermore, the clinical overlap between volume overload from missed dialysis and heart failure exacerbation may lead to missed or inaccurate diagnosis and ICD code misclassification.^{30,31} Even when stringent criteria are applied, adjudication of heart failure events in a clinical trial setting remains challenging in this population.³¹ Second, the magnitude and pace of GLP-1 RA–associated weight loss resulting from GLP-1 RA therapy may predispose patients to iatrogenic fluid overload if the estimated dry weight is not recalibrated in response. In standard, thrice-weekly in-center hemodialysis, the usual cadence of clinical reassessment may lag behind rapid changes in body composition. The loss of nonfluid body mass, without an appropriate downward adjustment of estimated dry weight, effectively replaces lost lean tissue with excess extracellular volume. Clinical trial protocols studying GLP-1 RA in kidney failure need to provide detailed criteria for heart failure adjudication and clear guidance for dry weight management to better elucidate the effect of GLP-1 RA on heart failure.

Furthermore, nephrologists need to carefully assess the volume status of patients receiving GLP-1 RA, particularly during drug escalation and periods of rapid weight loss.

The mechanisms by which GLP-1 RA exert cardiovascular benefit remain unclear, although proposed mechanisms include decreased BP, anti-inflammatory properties, favorable effects on cardiac output, and protection of endothelial function.^{27,32} Reduction in hemoglobin A1c is more pronounced with GLP-1 RA than sodium-glucose cotransporter 2 inhibitors, and secondary analyses of placebo-controlled trials suggest that this glycemic benefit may mediate a greater proportion of the cardiovascular benefits for GLP-1 RA than sodium-glucose cotransporter 2 inhibitors.³³ Although we did not monitor longitudinal changes in hemoglobin A1c, DPP4i and sulfonylureas generally provide similar glycemic control as compared with GLP-1 RA. Thus, glycemic control is unlikely to account for the suggested cardiovascular benefits observed in this study. While doses for weight loss exceed those used for glycemic control and the FLOW CKD outcomes trial,⁸ FLOW trial participants lost 4 kg more on GLP-1 RA than on placebo.⁸ Thus, weight loss may mediate some of the cardiovascular benefits of GLP-1 RA. Finally, given the substantial improvement in cardiovascular mortality in the FLOW trial, we infer that most of the association with improved all-cause mortality in this study reflects a reduction in cardiovascular mortality. However, prior comparative effectiveness data for GLP-1 RA in advanced CKD suggest that significant improvement in noncardiovascular mortality is also possible.²⁸

In our exploratory analyses, the association between GLP-1 RA and reduced risk of MACE was stronger in women. While GLP-1 RA use leads to greater weight loss in women,³⁴ this physiologic difference has not consistently translated into superior cardiovascular outcomes. In a comparative effectiveness study of GLP-1 RA use on a background of metformin in people with type 2 diabetes, GLP-1 RA use was more strongly associated with cardiovascular benefit in women.³⁵ However, a meta-analysis of placebo-controlled cardiovascular outcomes trials, which included the FLOW trial, found no evidence of sex-based differences in cardiovascular benefit.³⁶ Therefore, this exploratory finding warrants confirmation in dedicated prospective studies. In addition to sex, our study findings suggest that patients with a history of cardiovascular disease may derive superior MACE benefit than patients without cardiovascular disease. The waning nature of this association over the 2-year study period may reflect an early depletion of susceptible individuals (*i.e.*, higher event rate early in patients with cardiovascular disease) or biased medication discontinuation, resulting in a lower-risk cohort during later follow-up. These findings underscore the importance of these key subgroups in the design of prospective interventions in this population.

The observational design of this study introduces important limitations. Despite a substantial number of clinical covariates used to calculate propensity scores for these well-matched cohorts, no method short of randomization can entirely eliminate residual confounding. Select dialysis data were also not consistently available, including intact parathyroid hormone and serum phosphate levels. We did not adjudicate clinical outcomes in this study, which relied

on claims data to ascertain events. In addition, while USRDS is the largest collection of dialysis information in the United States, we were limited to patients with continuous Medicare Part D coverage and could not capture prescriptions filled through private insurance. In addition, we could not assess actual pharmacy dispensing or medication adherence. Although much of the study period preceded the rapid increase in GLP-1 RA use for weight loss (*e.g.*, semaglutide was approved for this indication in 2021), liraglutide was approved by the FDA for weight loss in 2014. Although we account for the year of prescription and obesity in the propensity modeling, some confounding by indication persists. Finally, the relatively short follow-up period limits any inference about the durability of the observed associations between GLP-1 RA and cardiovascular outcomes.

In a propensity score-matched cohort of patients with type 2 diabetes receiving maintenance dialysis, GLP-1 RA initiation was associated with a lower risk of cardiovascular outcomes and all-cause mortality compared with other antidiabetic agents. Given their strong safety profile and established cardiovascular benefits in nondialysis CKD, GLP-1 RA are a compelling option to improve cardiovascular outcomes for patients with type 2 diabetes and kidney failure, although randomized controlled trial data are necessary to confirm these findings and provide a clearer assessment of the potential risks and benefits in this population.

Disclosures

Disclosure forms, as provided by each author, are available with the online version of the article at <http://links.lww.com/JSN/F753>.

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these data are the responsibility of the author(s) and in no way should be seen as an official policy or interpretation of the US government. Christina Nix, Analyst Programmer with Duke University Department of Medicine, provided data management and statistical analysis assistance.

Declarative Statements

This study includes clinical experimentation and received Institutional Review Board or Ethics Committee approval. The need to obtain informed patient consent was waived.

Data Availability Statements

Data belong to a third party, and authors are not authorized to share the data. Identity of Third Party: USRDS. Reason for Restriction: Data were supplied by United States Renal Data System (USRDS). Deidentified patient data used in the project are available through request from USRDS.

Supplemental Material

This article contains the following supplemental material online at <http://links.lww.com/JSN/F754>.

Supplemental Table 1. List of ICD 9 and 10 codes for covariates and outcomes.

Supplemental Table 2. List of covariates and source within USRDS Files.

Supplemental Table 3. Baseline characteristics of the GLP-1 RA initiators as compared with sulfonylurea initiators before and after propensity score matching.

Supplemental Table 4. Cardiovascular outcomes for GLP-1 RA initiators as compared with DPP4i initiators and sulfonylurea initiators, modeling using subdistribution hazards to model competing risks.

Supplemental Table 5. Time-to-access intervention as a falsification end point for the primary and secondary analyses.

Supplemental Table 6. Distribution of the types of GLP-1 RA prescribed across the primary and secondary analyses.

Supplemental Figure 1. Subgroup analysis for the secondary outcomes in the GLP-1 RA versus DPP4i primary cohort.

Supplemental Figure 2. Subgroup analysis for all-cause mortality in the GLP-1 RA versus DPP4i primary cohort.

Supplemental Figure 3. Subgroup analysis for heart failure hospitalization in the GLP-1 RA versus DPP4i primary cohort.

Supplemental Figure 4. Consort diagram for selection of the secondary cohort of GLP-1 RA versus sulfonylurea initiators.

Supplemental Figure 5. Kaplan-Meier survival curve for MACE in the secondary cohort of GLP-1 RA versus sulfonylurea initiators.

Supplemental Figure 6. Subgroup analysis for the primary outcome in the secondary cohort of GLP-1 RA versus sulfonylurea initiators.

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